THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 1: **Exploring Substances: Acidic, Basic and Neutral** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A student mixes equal volumes of acid and base. The mixture still turns blue litmus red. This suggests: 1

(a) the acid was stronger or more concentrated
(b) the base was stronger
(c) the mixture is neutral
(d) no reaction

2. A cook adds a pinch of baking soda to lemon-flavoured drink and it fizzes. This shows: 1

(a) salt reaction **(b)** acid-carbonate reaction
(c) no reaction **(d)** sugar reaction

3. Which of the following reaction shows a salt being formed? 1

(a) $2\,Mg + O_2 \longrightarrow 2\,MgO$ **(b)** $C + O_2 \longrightarrow CO_2$ **(c)** $2\,H_2 + O_2 \longrightarrow 2\,H_2O$ **(d)** $HCl + NaOH \longrightarrow NaCl + H_2O$

4. A pH-like indicator turns green in a solution. A drop of unknown liquid Z turns it red. Z is: 1

(a) basic **(b)** neutral **(c)** salt **(d)** acidic

5. Which one of the following is both acidic in taste and used as a food preservative? 1

(a) lime water **(b)** soap **(c)** baking soda **(d)** vinegar

6. In which of the following will the red litmus turn blue? 1

(a) lemon **(b)** vinegar **(c)** aqueous $NaOH$ **(d)** aqueous HCl

7. Which of these will not show a colour change in turmeric? 1

(a) soap solution **(b)** washing soda **(c)** sugar solution **(d)** lime water

8. Which of these household items would best relieve the pain of a sting that contains formic acid? 1

(a) tamarind pulp **(b)** vinegar
(c) a paste of baking soda and water **(d)** lemon juice

9. Why can a lemon juice be used to remove the black layer on copper vessels? 1

(a) lemon is a metal **(b)** lemon is neutral
(c) the acid dissolves copper compounds **(d)** lemon is a base

10. Which of the following may be used as an antacid? 1

(a) hydrochloric acid **(b)** sulphuric acid
(c) vinegar **(d)** magnesium hydroxide

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does rubbing of baking soda paste help remove tarnish from silverware? 2

12. What could be used to neutralise a spilled acid in the lab safely? 2

13. Explain why pickles should not be stored in iron containers. 2

14. Why do aquatic animals die when acid rain falls in a lake? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain with equations how baking soda reacts with an acid in a cake batter to make it rise. 3

16. A factory discharges waste that turns blue litmus red. Suggest a treatment method. 3

17. Identify the odd one: vinegar, lemon, baking soda, curd. Give a reason. 3

OR

A person's teeth show decay. How are acids involved and what preventive steps are advised? 3

18. Explain how a baking soda solution can help stop the pain of an ant bite and how to check if the area is neutral. 3

OR

Explain what happens to turmeric stain on a white shirt when soap is applied and then lemon juice is added. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. You have vinegar, baking soda, lemon juice, soap solution and a few indicators. Plan a project to prepare a chart of indicators with colours and discuss limitations. 5

OR

Explain the working of a simple baking powder recipe and the acid-base reaction. 5

20. A test for acid rain is planned for your town. Describe steps, tools and conclusions. 5

OR

A student suspects that a drinking water sample is acidic. Plan the test and a safe next step. 5

END OF QUESTION PAPER — Total: 40 Marks

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